Spacing-Uniform Lattice Inequalities Cannot Produce the Yang–Mills Mass Gap

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We prove that a family of reflection-positive lattice gauge theories whose transfer matrices have a spectral gap bounded below in lattice units, uniformly in the spacing, has renormalised two-point functions whose limits vanish at every temporal separation after the first one at which a limit exists, so that it cannot converge to a continuum theory with a finite positive mass gap. Spacing-uniform log-Sobolev, Poincaré, Dobrushin and clustering bounds are therefore excluded as routes to the Yang–Mills existence and mass-gap problem. We prove three further obstructions: global Holley–Stroock perturbation of Haar measure by the Wilson weight loses the factor $e^{-2N\beta|P|}$; bounded nonnegative plaquette terms do not bound the free-energy cumulants uniformly in the volume, and plaquette variables satisfy exact Bianchi relations in dimension three or more; the one-loop effective coupling takes every positive value at a scale independent of the spacing, so no continuum limit avoids intermediate coupling. Every one of these statements is checked in Lean 4 with Mathlib, with every proof complete and no axiom beyond the three standard ones. We use them to locate the errors of eight manuscripts that claimed the mass gap by this route, we record what is known unconditionally, and we state the single residual statement that separates it from the Clay problem. The problem remains open.

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I. INTRODUCTION

The Yang–Mills existence and mass-gap problem [16] asks for a quantum SU(N) gauge theory on $\mathbb{R}^4$ satisfying the Wightman or Osterwalder–Schrader axioms [13, 24, 25] whose Hamiltonian has spectrum $\{0\} \cup [\Delta, \infty)$ with $\Delta > 0$. The standard starting point is Wilson's lattice theory [32]. For spacing $a > 0$ and coupling $\beta = 2N/g^2$ the Wilson measure on SU(N)-valued link variables is reflection positive [26], so it has a positive transfer matrix T_a and a lattice Hamiltonian $-\log T_a$; at strong coupling (small β) convergent expansions give a transfer-matrix gap and exponential clustering [17, 26, 29]. The unsolved content is the continuum limit $a \to 0$ along the asymptotic-freedom trajectory $\beta(a) \to \infty$.

Between 2025 and 2026 the authors circulated eight manuscripts (preprints 226502, 226913, 227159, 227445, 228638 and 229061, two of them in two copies) built on one strategy: a lower bound on the Bakry–Émery curvature of the gauge-orbit space [2, 21], transferred to the Wilson measure by Holley–Stroock perturbation [14] and Dobrushin–Stroock–Zegarlinski assembly [31], gives a log-Sobolev constant uniform in the volume and the spacing, hence a uniform lattice gap, hence a continuum mass gap. The last of the eight, 229061, already observed that this strategy defeats itself. The present paper replaces those manuscripts. It proves that observation in a stronger form, adds three further obstructions, checks all four in Lean, and lists the errors of the earlier texts by file and line.

Theorem 1 (No-go; Theorem 6)**.** *Let lattice theories indexed along any family of spacings $a \to 0^+$ have transfer-matrix gaps $\Delta_{\mathrm{lat}}(a) \geq \delta > 0$ in lattice units. For every choice of gauge-invariant observables and renormal-*

isations $Z_a > 0$, if the renormalised temporal two-point functions converge at separations $0 \le t_1 < t_2$, the limit at t_2 is zero. No such family converges to the two-point function of a continuum theory with gap $\Delta \in (0,\infty)$ and a nonzero spectral measure.

Theorem 2 (Volume obstruction; Proposition 13)**.** *For $U \in \mathrm{U}(N)$, $|\mathrm{Re}\,\mathrm{tr}\,U| \le N$. The Wilson action over a set P of plaquettes has oscillation at most $2N|P|$, and the Holley–Stroock factor $e^{-2N\beta|P|}$ tends to zero as $|P| \to \infty$ at every $\beta > 0$.*

Theorem 3 (Cumulants and Bianchi; Section V)**.** *There are actions that are sums of n local terms in $[0, 2N]$ with $\mathrm{Var}(S_n)/n = N^2 n$. On every unit cube the six oriented plaquette angles of an abelian configuration sum to zero.*

Theorem 4 (The window is entered; Section VI)**.** *The one-loop coupling $g^2(\mu) = 1/(2b_0 \log(\mu/\Lambda))$, $b_0 = 11N/(48\pi^2)$, takes every value $g_*^2 > 0$ at $\mu_* = \Lambda \exp(1/(2b_0 g_*^2))$, a scale independent of a and inside $(\Lambda, 1/a)$ for every $a < 1/\mu_*$.*

Theorems 1 to 4 are checked in Lean 4, as described in Section XII. Together they show where each of the earlier manuscripts fails (Section VIII): the curvature step proves too little (it controls only Haar measure, or strong coupling), the analyticity step uses an independence that is false in four dimensions, the coupling step assumes that the continuum limit avoids intermediate coupling, which is impossible, and the final step, a gap uniform in the spacing, would refute the goal it was meant to prove.

Section IX records what is known unconditionally, and Section X examines the other claimed proofs we found. Section XI states the residual statement. The paper proves no existence or gap statement for the continuum theory.

II. SETTING

Fix $N \ge 2$. For a finite hypercubic lattice $\Lambda \subset a\mathbb{Z}^4$ with links E and plaquettes P, the Wilson measure is

$$d\mu_{\Lambda,\beta}(U) = Z^{-1} e^{-\beta S(U)} \prod_{e \in E} dU_e,$$
$$S(U) = \sum_{p \in P} \big(N - \mathrm{Re}\,\mathrm{tr}\,U_p\big), \tag{1}$$

with dU_e normalised Haar measure and U_p the ordered product of links around p. Reflection positivity in temporal planes [26] gives a Hilbert space $\mathcal{H}_a$, a self-adjoint transfer matrix $0 \le T_a \le 1$ with ground state Ω_a, and for a gauge-invariant observable $\mathcal{O}_a$ localised in a time slice the spectral representation of the truncated temporal two-point function at $t \in a\mathbb{Z}_{>0}$:

$$S_a(t) = \int_{[0,\infty)} e^{-Et/a}\, d\rho_a(E), \tag{2}$$

where E runs over the spectrum of $-\log T_a$ (lattice units) on the orthogonal complement of Ω_a and ρ_a is the spectral measure of $(\mathcal{O}_a - \langle \mathcal{O}_a \rangle)\Omega_a$, a finite positive measure. A lattice gap $\Delta_{\mathrm{lat}}(a) \ge \delta$ means $\rho_a([0,\delta)) = 0$ for all such observables. Formula (2) defines $S_a(t)$ for every real $t \ge 0$; along any sequence of spacings for which $t \in a\mathbb{Z}$ it is the lattice two-point function, and the theorems below are stated along an arbitrary filter so that they apply to every such sequence.

The physical gap of the lattice theory is $\Delta_{\mathrm{lat}}(a)/a$. A continuum theory with mass m requires $\Delta_{\mathrm{lat}}(a) = ma + o(a)$; the earlier manuscripts wrote Δ_a for $\Delta_{\mathrm{lat}}(a)$ and then treated it alternately as a lattice-unit and a physical quantity (Section VIII).

III. THE NO-GO THEOREM

Lemma 5 (Ratio bound)**.** *If ρ is a finite positive measure with $\rho([0,\delta)) = 0$, $\delta \ge 0$, $a > 0$ and $0 \le t_1 \le t_2$, then*

$$S_a(t_2) \le e^{-\delta(t_2 - t_1)/a}\, S_a(t_1). \tag{3}$$

Proof. On the support $E \ge \delta$, so $e^{-Et_2/a} = e^{-E(t_2-t_1)/a} e^{-Et_1/a} \le e^{-\delta(t_2-t_1)/a} e^{-Et_1/a}$; both integrands are bounded by 1 on the support, hence integrable for the finite measure ρ; integrate. □

Every renormalisation constant cancels in (3), and its right side carries the factor $e^{-c/a}$, $c = \delta(t_2 - t_1) > 0$.

Theorem 6 (Spacing-uniform gaps force ultralocal limits)**.** *Let (a_i) tend to 0^+ along a filter, let ρ_i be finite positive measures with $\rho_i([0,\delta)) = 0$ for a fixed $\delta > 0$, and let $Z_i > 0$. If $Z_i S_{a_i}(t_1) \to L_1$ and $Z_i S_{a_i}(t_2) \to L_2$ with $0 \le t_1 < t_2$, then $L_2 = 0$.*

Proof. By (3), $0 \le Z_i S_{a_i}(t_2) \le e^{-c/a_i}\, Z_i S_{a_i}(t_1)$. The first factor tends to 0 and the second to L_1, so the middle term tends to 0. □

The statement is stronger than the one in 229061, which assumed $L_1 \ne 0$: convergence at the earlier time alone forces the later limit to vanish.

Lemma 7 (Källén–Lehmann positivity)**.** *If $\sigma \ne 0$ is a finite positive measure with $\sigma([0,\Delta)) = 0$, $\Delta > 0$, then $S(t) = \int e^{-Et} d\sigma(E) > 0$ for every $t > 0$.*

Proof. The integrand is strictly positive and bounded by 1 on the support, and $\sigma(\mathbb{R}) > 0$. □

Corollary 8. *Under the hypotheses of Theorem 6, the renormalised two-point functions cannot converge at every $t > 0$ to the two-point function of a continuum theory with gap $\Delta > 0$ and nonzero spectral measure. If $\Delta_{\mathrm{lat}}(a)/a \to m$, then $\Delta_{\mathrm{lat}}(a) \to 0$, so no $\delta > 0$ bounds Δ_{lat} below.*

Proof. Apply Theorem 6 with $t_1 = 1$, $t_2 = 2$: the limit at $t = 2$ would be 0, against Lemma 7. For the second statement, $\Delta_{\mathrm{lat}}(a) = (\Delta_{\mathrm{lat}}(a)/a)\, a \to m \cdot 0$. □

Lemma 9 (Critical scaling). *Let (a_i) tend to 0^+ along a filter, let ρ_i be finite positive measures with $\rho_i([0, m_i a_i)) = 0$ for numbers $m_i \geq 0$ with $m_i \to m$, and let $Z_i > 0$. If $Z_i S_{a_i}(t_1) \to L_1$ and $Z_i S_{a_i}(t_2) \to L_2$ with $0 \leq t_1 \leq t_2$, then $L_2 \leq e^{-m(t_2-t_1)} L_1$.*

Proof. Lemma 5 with $\delta = m_i a_i$ gives $Z_i S_{a_i}(t_2) \leq e^{-m_i(t_2-t_1)} Z_i S_{a_i}(t_1)$; the right side tends to $e^{-m(t_2-t_1)} L_1$. □

Lemma 9 is the bound that Theorem 6 leaves open. A lattice gap bounded below by a constant is excluded; a lattice gap bounded below by ma makes every limiting two-point function satisfy the ratio bound of a theory with gap m, and a Laplace transform of a finite positive measure obeying $S(t_2) \leq e^{-m(t_2-t_1)} S(t_1)$ for all $t_1 \leq t_2$ has its measure supported in $[m, \infty)$. The converse does not follow from anything here: a gapped limit forces $\Delta_{\mathrm{lat}}(a) \to 0$ (Corollary 8) and $\limsup \Delta_{\mathrm{lat}}(a)/a \leq m$ when the gap m is attained, but the lower bound $\liminf \Delta_{\mathrm{lat}}(a)/a \geq m$ is exactly what has to be proved, since the gap is only lower semicontinuous under convergence of two-point functions.

Corollary 10 (Foreclosed inequalities). *Along a family of Wilson theories converging after renormalisation to a continuum theory with gap in $(0, \infty)$, none of the following holds with constants uniform in a: a log-Sobolev or Poincaré inequality for μ_a in the lattice-unit Dirichlet form; a Dobrushin–Stroock–Zegarlinski mixing condition; exponential clustering of gauge-invariant local observables at a lattice-unit rate.*

Proof. For finite-range Gibbs measures, uniform log-Sobolev and Dobrushin–Stroock–Zegarlinski conditions imply exponential decay of correlations at a lattice-unit rate [31], and a uniform Poincaré inequality for the infinite-volume measure implies the same for local observables. Clustering in the time direction at rate δ for a dense set of local observables gives $\rho_a([0, \delta)) = 0$ through (2), which is the hypothesis of Theorem 6. □

The formal proof covers Lemma 5, Theorem 6, Lemma 7 and Corollary 8; the implications from functional inequalities to clustering in Corollary 10 are quoted.

Figure 1 draws the ratio bound for three spacings.

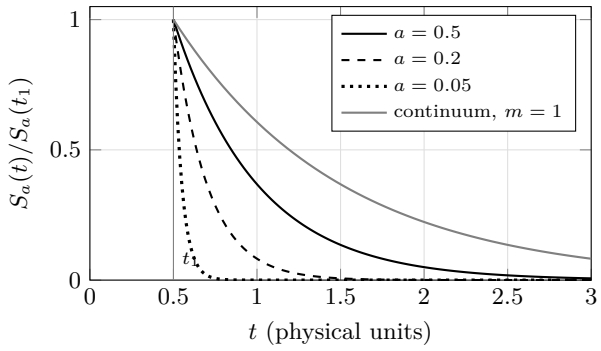


FIG. 1. The bound $e^{-\delta(t-t_1)/a}$ of Lemma 5 with $\delta = 1$, for three spacings, against a continuum two-point function of mass 1. As $a \to 0$ the bound collapses onto the vertical line $t = t_1$, whatever the renormalisation; a gapped continuum limit needs $\delta = \delta(a) = ma$.

IV. THE VOLUME OBSTRUCTION

Lemma 11. *For $U \in \mathrm{U}(N)$, $|\mathrm{Re}\,\mathrm{tr}\, U| \leq |\mathrm{tr}\, U| \leq N$. Hence $0 \leq N - \mathrm{Re}\,\mathrm{tr}\, U \leq 2N$, $0 \leq S(U) \leq 2N|P|$ in (1), and $\beta S(U) - \beta S(U') \leq 2N\beta|P|$ for $\beta \geq 0$.*

Proof. Each diagonal entry of a unitary matrix has modulus at most one; sum the diagonal. The product of the links around a plaquette is unitary. □

The bound $2N$ on the oscillation of a single plaquette term is attained for even N (at $U = -I \in \mathrm{SU}(N)$); for odd N the minimum of $\mathrm{Re}\,\mathrm{tr}$ on $\mathrm{SU}(N)$ is $-N\cos(\pi/N)$, so the oscillation is $N(1 + \cos(\pi/N)) < 2N$. Proposition 12 is the Holley–Stroock lemma, proved here on a finite state space in both its Poincaré and its entropy form; the argument for a compact group is the same with sums replaced by integrals [14].

Proposition 12 (Holley–Stroock). *Let ν be a probability vector on a finite set, $m \leq V \leq M$, and $\mu = \nu e^{-V}/Z$. Let $\Gamma \geq 0$ be an energy density. If $\mathrm{Var}_\nu(f) \leq C \sum \nu\Gamma$ then $\mathrm{Var}_\mu(f) \leq C e^{M-m} \sum \mu\Gamma$. If $g \geq 0$ and $\mathrm{Ent}_\nu(g) \leq C \sum \nu\Gamma$ then $\mathrm{Ent}_\mu(g) \leq C e^{M-m} \sum \mu\Gamma$.*

Proof. For a probability vector w, $\mathrm{Var}_w(f) = \min_c \sum w(f-c)^2$ and, for $g \geq 0$, $\mathrm{Ent}_w(g) \leq \sum w\, \phi(g, c)$ for every $c > 0$, with equality at $c = \sum wg$ when that mean is positive, where $\phi(x, c) = x\log x - x\log c - x + c \geq 0$. Bound the μ-average of the nonnegative integrand by e^{-m}/Z times its ν-average, apply the hypothesis, and bound $\sum \nu\Gamma \leq e^M Z \sum \mu\Gamma$. If the ν-mean of g vanishes then $g = 0$ on the support of ν, hence of μ, and both entropies vanish. □

Proposition 13 (Volume obstruction). *Perturbing the product Haar measure on $\mathrm{SU}(N)^E$, which satisfies a log-Sobolev inequality with a constant α_N independent of E, by the full Wilson weight gives the constant $\alpha_N e^{-2N\beta|P|}$, which tends to 0 as $|P| \to \infty$ for every $\beta > 0$.*

Proof. Proposition 12 with $V = \beta S$ and Lemma 11; the limit is $\exp(-2N\beta m) \to 0$ as $m \to \infty$. □

Volume-uniform constants require the local transfer of [31] or the Bakry–Émery criterion applied to the interacting measure itself (Section VII); both hold at strong coupling only.

V. FREE-ENERGY CUMULANTS AND THE BIANCHI IDENTITY

For $f_\Lambda(\beta) = -|\Lambda|^{-1} \log \int e^{-\beta S} \prod dU_e$,

$$f'_\Lambda(\beta) = |\Lambda|^{-1} \langle S \rangle_\beta, \qquad f''_\Lambda(\beta) = -|\Lambda|^{-1} \operatorname{Var}_\beta(S), \quad (4)$$

and in general $f^{(n)}_\Lambda(\beta) = (-1)^{n+1} |\Lambda|^{-1} \kappa_n(S)$ with κ_n the cumulants of S under $\mu_{\Lambda,\beta}$. A bound $|f^{(n)}_\Lambda| \le n!\, C^n$ uniform in Λ is therefore the statement $\kappa_n(S) = O(|\Lambda|)$, which is a decorrelation property of the measure, and is exactly what cluster expansions prove at strong coupling.

Proposition 14. *Let $Y \in \{0,1\}$ be a fair coin and $S_n = \sum_{i=1}^{n} 2NY$. Each term lies in $[0, 2N]$ and $\operatorname{Var}(S_n)/n = N^2 n$, which is unbounded in n.*

Proof. $\langle S_n \rangle = Nn$ and $\operatorname{Var}(S_n) = \frac{1}{2}(Nn)^2 + \frac{1}{2}(Nn)^2$. □

Boundedness and nonnegativity of the local terms, the only properties used in the analyticity arguments of the earlier manuscripts, therefore do not bound the cumulants. Those arguments also asserted that the plaquette variables are independent under Haar measure. They are not, in any dimension $d \ge 3$:

Proposition 15 (Bianchi identity on a cube)**.** *Let θ_e take values in an abelian group, and for a plaquette with boundary edges e_1, e_2, e_3, e_4 let $\theta_p = \theta_{e_1} + \theta_{e_2} - \theta_{e_3} - \theta_{e_4}$. On a unit cube the six oriented plaquette angles sum to zero identically.*

For U(1), or for the maximal torus of SU(N), the six plaquette angles of a cube are thus linearly dependent, so they cannot be independent uniform variables under Haar measure on links; for SU(N) the non-abelian Bianchi identity makes a conjugated product of the six plaquette holonomies equal to the identity. The analyticity of the four-dimensional Wilson free energy on $(0, \infty)$ is an open problem.

VI. RUNNING COUPLING AND THE INTERMEDIATE WINDOW

The one-loop solution of $\mu\, dg/d\mu = -b_0 g^3$ with $b_0 = 11N/(48\pi^2)$ is $g^2(\mu) = 1/(2b_0 \log(\mu/\Lambda))$ for $\mu > \Lambda$. It tends to 0 as $\mu \to \infty$ and takes the value g_*^2 at $\mu_* = \Lambda e^{1/(2b_0 g_*^2)}$, which lies in $(\Lambda, 1/a)$ whenever $a < 1/\mu_*$. Along any continuum trajectory the bare coupling $g^2(1/a)$ tends to zero, but the effective coupling at the fixed physical scale μ_* equals g_*^2 at every small spacing. A renormalisation-group analysis that integrates out scales from $1/a$ down to the mass scale must therefore pass through every intermediate coupling. The claim of 226913, 227445 and 228638 that "the continuum limit lies entirely inside the Balaban regime and the intermediate coupling window is never entered" confuses the bare coupling with the effective coupling at physical scales. Figure 2 draws the curve.

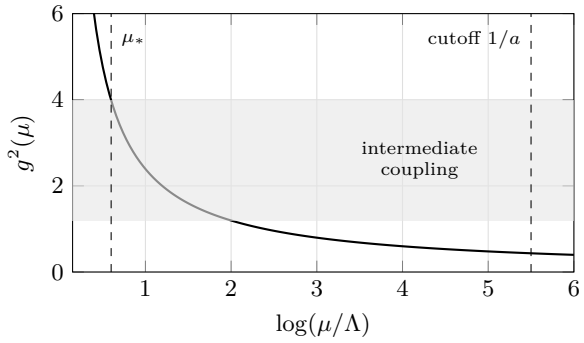


FIG. 2. One-loop running coupling for $N = 2$ ($2b_0 \approx 0.418$). The bare coupling at the cutoff is small, but between the cutoff and Λ the curve crosses the shaded band at a scale μ_* that does not move as $a \to 0$.

VII. CURVATURE OF THE GAUGE-ORBIT SPACE

Lemma 16 (Milnor)**.** *With the bi-invariant metric $\langle X, Y \rangle = -2 \operatorname{tr}(XY)$ on* SU(N)*, $\mathrm{Ric} = \frac{N}{4}\, g$.*

This is correct as stated in 227159 (lines 369–393): for a bi-invariant metric $\mathrm{Ric}(X,X) = -\frac{1}{4} B(X,X)$, and the Killing form of $\mathfrak{su}(N)$ is $B(X,X) = 2N \operatorname{tr} X^2$. The Ricci tensor of $\mathrm{SU}(N)^E$ is $\frac{N}{4} g$ for every E, so by the Bakry–Émery criterion the product Haar measure satisfies the log-Sobolev inequality

$$\operatorname{Ent}(f^2) \le \frac{8}{N} \int |\nabla f|^2 \, d\mathrm{Haar} \quad (5)$$

for every smooth f on $\mathrm{SU}(N)^E$, uniformly in E. Gauge-invariant functions are functions on $\mathrm{SU}(N)^E$, so (5) already contains every statement about them that the orbit space could give; the passage to $\mathcal{B} = \mathcal{A}/\mathcal{G}$ adds nothing.

The passage as written is also incorrect. The earlier manuscripts claim that every gauge orbit is a minimal submanifold of $\mathrm{SU}(N)^E$ (227159, Lemma 16's successor at lines 413–441; 227445 line 499; 228638 line 774; 226913 line 1398).

Proposition 17. *On the one-site periodic lattice every link is a loop and the gauge group acts by simultaneous conjugation. For* SU(2) *with the round metric on S^3, the orbit of a configuration $(U_1, I, \dots, I)$ with $\operatorname{tr} U_1 = 2\cos\theta$, $0 < \theta < \pi$, is a geodesic sphere of radius θ about the identity in the first factor, with mean curvature $2\cot\theta$. It is minimal only for $\theta = \pi/2$.*

Proof. Identify SU(2) with the unit quaternions. The conjugacy class of $\cos\theta + u \sin\theta$, u a unit imaginary quaternion, is $\{\cos\theta + v\sin\theta : |v| = 1,\ v \text{ imaginary}\}$, the set of points at geodesic distance θ from 1, a round two-sphere of radius $\sin\theta$ whose principal curvatures in S^3 are $\cot\theta$. The other factors are fixed points of conjugation. □

The proof in 227159 fails at two points. On a loop link $s(e) = t(e) = x$ the fundamental field is $T_a U_e - U_e T_a$, which is neither left- nor right-invariant, so $\nabla_X X \neq 0$; and the mean curvature is the trace of the second fundamental form over an orthonormal frame of the orbit, not the sum of $\nabla_{X_{a,x}} X_{a,x}$ over the fundamental fields, whose lengths vary along the configuration space. For an isometric action the mean curvature of a principal orbit is the gradient of $-\log$ of the orbit volume, which is not constant. What is true is the weighted statement: the quotient of a space with Ricci curvature at least K by a compact group of isometries, with the push-forward of the volume measure, satisfies the curvature-dimension condition $\mathrm{CD}(K,\infty)$ [12, 19]. With $K = N/4$ this is the correct form of the curvature theorem of the earlier manuscripts; it is a statement about Haar measure and is equivalent in content to (5) for gauge-invariant functions.

For the Wilson measure itself, the Bakry–Émery criterion needs $\mathrm{Ric} + \beta \operatorname{Hess} S \geq K > 0$. The Hessian of S is bounded in operator norm by a constant times $d-1$ per link, so this holds for β below a threshold of order $1/(N(d-1))$. Shen, Zhu and Zhu proved it in this form, with 't Hooft scaling and the threshold $|\beta| < 1/(16(d-1))$ for $\mathrm{SU}(N)$, and derived from it the log-Sobolev inequality, uniqueness of the infinite-volume measure and a mass gap [30]. That is a strong-coupling theorem: it holds at fixed β, that is at fixed lattice coupling, and by Corollary 10 it cannot hold with constants uniform along the continuum trajectory.

VIII. AUDIT OF THE EIGHT MANUSCRIPTS

The manuscripts are 226502 (two identical copies), 226913, 227159, 227445 (two identical copies), 228638 and 229061. Line numbers refer to the source files. Table I summarises; the errors are as follows.

(E1) Uniform lattice gap. 226502 (abstract, lines 112–121, and Theorem lattice-gap), 226913 (lines 2256, 2309), 227445 (line 139) and 228638 (lines 216, 1679) assert $\Delta_{\mathrm{lat}}(a) \geq \alpha_*/2$ uniformly in a. By Theorem 6 this excludes a continuum limit with a finite gap. The same manuscripts then take $\Delta_0 = \lim a\Delta_{\mathrm{lat}}(a)$ (226502 line 121) or $\Delta_0 \geq \limsup \Delta_{\mathrm{lat}}(a)$ (227445 line 1016), mixing lattice and physical units.

(E2) Internal contradiction in the physical gap. In 226502, lines 1793–1845, the proof first obtains $a\Delta_{\mathrm{lat}}(a) \geq \frac{N}{8}(1-\varepsilon) C_{\mathrm{RS}} \Lambda^{-1} e^{-4\pi^2\beta/N}$, which tends to 0, and then inserts the bound $\alpha_* \geq N\kappa_{\min}^2/(4 g_{\mathrm{phys}}^2 a^2)$, which tends to ∞ as $a \to 0$, against the definition $\alpha_* = (N/4)(1-\varepsilon(\beta)) \leq N/4$ used in the same proof. The Hessian of $-\mathrm{Re}\,\mathrm{tr}\, U_p$ in the link variables does not scale like a^{-2}.

(E3) Spectral transfer. 227445 lines 1000–1016 and 228638 line 1864 apply the Trotter–Kato theorem to operators H_a that act on different Hilbert spaces $\mathcal{H}_a$ without identifying them, and the lattice-unit gap transported by strong resolvent convergence [27] becomes, in physical units, $\Delta_{\mathrm{lat}}(a)/a \to \infty$: the spectrum of the limit is contained in $\{0\}$, which is the ultralocal conclusion of Theorem 6.

(E4) Orbit minimality. 227159 lines 413–441, 227445 line 499, 228638 line 774 and 226913 line 1398 assert that every gauge orbit is minimal; Proposition 17 is a counterexample. The conclusion survives only in the weighted form of Section VII, for Haar measure.

(E5) Sharpness. 227159 line 523 states equality in $\Gamma_2 \geq \frac{N}{4}\Gamma$ for affine-linear f on S^3. For the restriction f of a linear function to the round S^3, $\operatorname{Hess} f = -fg$ and $\Gamma_2(f) = 3f^2 + 2|\nabla f|^2$ in the unit normalisation, which is not $2|\nabla f|^2$; the bound is sharp as a spectral-gap estimate (Lichnerowicz), not pointwise.

(E6) Analyticity of the free energy. 227159 lines 771–799 (Lemma taylor), 226502 lines 105, 193, 2204–2249, 227445 line 799 and 228638 line 1459 bound $f_\Lambda^{(n)}$ uniformly in Λ by treating $|\Lambda|^{-1}S$ as a sum of independent terms under Haar measure. Three steps fail: the derivative at β_0 is a cumulant under μ_{Λ,β_0}, not under Haar measure (4); the cumulant of the normalised sum $|\Lambda|^{-1}S$ is $|\Lambda|^{-n}\kappa_n(S)$, so a bound on it does not bound $|\Lambda|^{-1}\kappa_n(S)$; and the plaquette variables are not independent (Proposition 15). Without independence the claim is false in general (Proposition 14). A Cauchy estimate on $\log M$ also needs $M \neq 0$ on a complex disc, which is not shown. The same argument, applied verbatim to the two-dimensional Ising model, whose bond terms $1 - \sigma_x\sigma_y$ are bounded and nonnegative, would prove analyticity of its free energy at the critical point, against Onsager's solution. The conclusion "no bulk phase transition at any $\beta > 0$ and every $N \geq 2$" is unproved; for the Wilson action at large N numerical studies report bulk transitions.

(E7) From analyticity to the Kotecký–Preiss criterion. 226502 lines 106 and 228–234 infer convergence of polymer expansions at every scale from analyticity of f by "Cauchy-estimate inversion". The Kotecký–Preiss criterion is a bound on polymer activities, which implies analyticity and clustering; there is no converse, and analyticity of the free energy does not imply exponential clustering.

(E8) Avoiding intermediate coupling. 226913 line 114, 227445 line 134 and 1124, 228638 line 1975: refuted by Theorem 4, Section VI.

(E9) Uniform log-Sobolev constant. $\alpha_* = (N/4)(1-\varepsilon(\beta))$ uniform in volume and spacing (226502 abstract; 227445 lines 127–130) is obtained by Holley–Stroock, whose constant is $e^{-2N\beta|P|}$ (Proposition 13); 227159 records this volume dependence correctly in its own Corollary lsi (line 640). The step from finite volume to a uniform constant is the Dobrushin–Stroock–Zegarlinski assembly, whose hypotheses hold at strong coupling only, and by Corollary 10 cannot hold uniformly along the continuum trajectory.

(E10) Remarks on 229061. Its no-go theorem is correct, and is Theorem 6 in a weaker form. Two statements need correction: the oscillation $2N\beta|P|$ is an upper bound, attained only for even N (Section IV); and the

TABLE I. Verdicts on the main claims of the eight manuscripts.

Claim	Where	Verdict
$\mathrm{Ric}_{\mathrm{SU}(N)} = \frac{N}{4} g$	227159:369	true (Milnor)
orbits minimal	227159:413	false (Prop. 17)
$\mathrm{CD}(\frac{N}{4}, \infty)$ on $\mathcal{B}$	227159:500	true weighted [12]
LSI $\frac{N}{4} e^{-2N\beta\lvert P\rvert}$	227159:640	true, not uniform
uniform LSI α_*	226502 abs.	unproved; (E9)
f analytic on $(0, \infty)$	227159:886	unproved; (E6)
analyticity $\Rightarrow$ KP	226502:233	false implication
window never entered	226913:114	false (Thm. 4)
$\Delta_{\mathrm{lat}} \geq \alpha_*/2$ uniform	226502:115	excluded (Thm. 1)
$\Delta_0 > 0$ in continuum	all but 229061	not proved
no-go	229061	true (Thm. 1)

asserted equivalence of $\Delta_{\mathrm{lat}}(a) = ma + o(a)$ with a log-Sobolev constant vanishing like a^2 is not a theorem, since the log-Sobolev constant is bounded by the spectral gap but no reverse bound uniform in the volume is known.

IX. WHAT IS KNOWN UNCONDITIONALLY

Reflection positivity of the Wilson measure for every compact gauge group and every coupling, with its transfer-matrix formalism, is due to Osterwalder and Seiler [26, 29]. A positive transfer-matrix gap and exponential clustering at strong coupling follow from convergent character and cluster expansions [17, 26, 29]; Shen, Zhu and Zhu obtained the log-Sobolev inequality, uniqueness and a mass gap at strong coupling from the Bakry–Émery criterion and Langevin dynamics [30], and Nissim extended these results to $\mathrm{U}(N)$ in the 't Hooft regime [23]. Chatterjee proved that exponential decay of correlations under arbitrary boundary conditions, for a gauge group with nontrivial centre, implies confinement [8]; Adhikari and Cao proved correlation decay for finite gauge groups at weak coupling [1]; Cao, Nissim and Sheffield enlarged the regime in which the area law is proved [5].

In the continuum, Bałaban's multiscale analysis proved ultraviolet stability of four-dimensional lattice gauge theories: bounds on effective actions and partition functions uniform in the spacing, for small bare coupling, in a fixed physical volume [3, 4]. Magnen, Rivasseau and Sénéor constructed four-dimensional Yang–Mills theory in finite volume with an infrared cutoff [20]. In two dimensions the continuum Yang–Mills measure is constructed [18], and its Langevin dynamics has been built [6]; in three dimensions Chandra, Chevyrev, Hairer and Shen constructed the stochastic quantisation of Yang–Mills–Higgs locally in time [7]. None of these constructs a four-dimensional measure on $\mathbb{R}^4$ satisfying the Osterwalder–Schrader axioms, and none proves a gap for one.

TABLE II. Claimed proofs examined, with their status.

Text	Status
Dynin, arXiv:0903.4727 and sequels	continuum quantisation with quartic term; Poincaré covariance and locality not established [9]
Farinelli, arXiv:1406.4177	Coulomb-gauge construction, seventeen versions, math.GM; not accepted [10]
Jacobsen, arXiv:2506.00284	withdrawn by arXiv administrators [15]
Franchi Viceré, Zenodo 17966583	finite-lattice gap by Perron–Frobenius; reduces the problem to "persistence of the gap", which in lattice units is excluded by Theorem 1 [11]
Nair, arXiv:2608.10133	outline, three dimensions; not four-dimensional [22]

X. OTHER CLAIMED PROOFS

We searched arXiv, Zenodo and PhilArchive for texts claiming the four-dimensional mass gap. Table II lists those we examined, with the mechanism stated in each abstract. We did not read every such text in full, and the list is not exhaustive. Theorem 1 applies to any argument whose last step passes from a lattice gap uniform in lattice units to a continuum gap; it neither repairs nor bears on arguments that work directly in the continuum, whose gaps lie elsewhere.

XI. THE RESIDUAL STATEMENT

Hypothesis 18 (Residual statement)**.** There exist $m \in (0, \infty)$, a trajectory $\beta = \beta(a) \to \infty$ and renormalisations of the gauge-invariant fields such that: (R1) the renormalised Schwinger functions of the Wilson theory on $a\mathbb{Z}^4$ converge as $a \to 0$ to limits satisfying OS1–OS5; (R2) $\liminf_{a \to 0} \Delta_{\mathrm{lat}}(a)/a \geq m$; (R3) the limiting two-point functions of some gauge-invariant field are nonzero.

By Lemma 9, (R1) and (R2) make every limiting two-point function the Laplace transform of a measure supported in $[m, \infty)$, so by Osterwalder–Schrader reconstruction Hypothesis 18 gives the Clay statement for the reconstructed theory, with gap at least m and, by (R3), a nontrivial theory. Theorem 6 shows that (R2) cannot be strengthened to a bound uniform in lattice units, and a bound by $o(a)$ gives no gap; (R2) is the one-sided bound at the critical scaling, and it is the Clay problem restated on the lattice. Relative to Bałaban's programme the missing parts are the following. Bałaban's bounds concern partition functions and effective actions in a fixed physical volume at small bare coupling; (R1) needs the convergence of correlation functions and the removal of the volume cutoff, which requires control of the renormalisation-group flow at the scales where the

effective coupling is of order one (Section VI). (R2) is a statement about exactly those scales: the lattice correlation length $1/\Delta_{\mathrm{lat}}(a)$ must diverge, by Corollary 8, but no faster than $1/(ma)$. No method now available controls the flow there in four dimensions.

The comparison with the Poincaré conjecture is instructive. Hamilton's Ricci flow reduced that problem to the analysis of singularities, and Perelman's noncollapsing and canonical-neighbourhood theorems classified them, so that surgery could continue the flow. Here the renormalisation-group flow is the analogue of the Ricci flow, Bałaban's theorems control it where the coupling is small, and the region of order-one coupling is the analogue of the singular set. A theorem that classified the effective actions in that region, as Perelman's classified the singular models, would be the step that is missing. Theorem 1 shows that it cannot be replaced by any inequality that is uniform in the spacing: whatever controls that region must produce the divergence of the correlation length, not exclude it.

XII. FORMAL VERIFICATION

The finite statements of this paper are checked in Lean 4 on Mathlib, in the archive [28]. They are Lemma 5, Theorem 6, Lemma 7, Corollary 8 and Lemma 9; Lemma 11, the bounds $0 \le S \le 2N|P|$ and the oscillation bound $2N\beta|P|$; Proposition 13; the Bianchi identity of Proposition 15; the variational forms of variance and entropy and both forms of Proposition 12; Theorem 4 with asymptotic freedom of the one-loop coupling; and Proposition 14. No proof is left incomplete, and each uses only the three standard axioms of the Lean kernel (propositional extensionality, choice and quotient soundness).

The formal statements are the abstract forms given above. The spectral representation (2) is a hypothesis of the formal no-go theorem, not a formal theorem: it is the spectral theorem for the transfer matrix, which we quote from [13, 26]. Likewise Milnor's formula, the Bakry–Émery criterion, Proposition 17, the quotient theorem of [12] and the implications of Corollary 10 are proved or quoted in the text and are not formalised; Riemannian geometry of Lie groups is not yet available in Mathlib. Holley–Stroock is formalised on a finite state space.

XIII. CONCLUSION

The strategy of the eight earlier manuscripts cannot reach the Yang–Mills mass gap, for a reason independent of the quality of its estimates: its final inequality, a gap uniform in the lattice spacing, contradicts the existence of a continuum limit with finite mass. Its intermediate steps contain the errors (E1)–(E10). What it does prove correctly, a curvature bound for Haar measure and a volume-dependent log-Sobolev constant for the Wilson measure, is classical, and its correct strong-coupling form is the theorem of Shen, Zhu and Zhu. The residual statement, Hypothesis 18, a lattice gap of at least ma together with convergence, is the Clay problem on the lattice; it requires control of the renormalisation-group flow at order-one coupling, and it remains open.

DATA AVAILABILITY

The Lean 4 source, the build configuration and the sources of this paper are archived on Zenodo as release v1.0.0, doi:10.5281/zenodo.23444338 [28]. The Lean formalisation was prepared with the assistance of AI tools and checked by the Lean kernel.

CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest and received no funding for this work.

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